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In [5]: import numpy as np

# Step function (activation function)
def step_function(x):
    if x >= 0:
        return 1
    else:
        return 0

# Training data (Example: AND gate)
X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])

# Expected output
y = np.array([0, 0, 0, 1])

# Initialize weights and bias
weights = np.zeros(2)
bias = 0
learning_rate = 0.1
epochs = 10

# Training the perceptron
for epoch in range(epochs):
    print("Epoch:", epoch+1)

    for i in range(len(X)):
        y_pred = step_function(np.dot(X[i], weights) + bias)

        # Calculate error
        error = y[i] - y_pred

        # Update weights and bias
        weights += learning_rate * error * X[i]
        bias += learning_rate * error

        print("Input:", X[i], "Predicted:", y_pred, "Actual:", y[i])

print("\nFinal Weights:", weights)
print("Final Bias:", bias)
```

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Epoch: 1
Input: [0 0] Predicted: 1 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 0 Actual: 1
Epoch: 2
Input: [0 0] Predicted: 1 Actual: 0
Input: [0 1] Predicted: 1 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 0 Actual: 1
Epoch: 3
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 1 Actual: 0
Input: [1 0] Predicted: 1 Actual: 0
Input: [1 1] Predicted: 0 Actual: 1
Epoch: 4
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 1 Actual: 1
Epoch: 5
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 1 Actual: 1
Epoch: 6
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 1 Actual: 1
Epoch: 7
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 1 Actual: 1
Epoch: 8
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 1 Actual: 1
Epoch: 9
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 1 Actual: 1
Epoch: 10
Input: [0 0] Predicted: 0 Actual: 0
Input: [0 1] Predicted: 0 Actual: 0
Input: [1 0] Predicted: 0 Actual: 0
Input: [1 1] Predicted: 1 Actual: 1

Final Weights: [0.2 0.1]
Final Bias: -0.20000000000000004
```